

Hide menu

Week 3: Image Sensing

- Reading: Week 3 Lecture Handout
30 min
- Ungraded Plugin: 3.1
Overview of Imaging Sensing
3 min
- Practice Quiz: 3.1
Overview of Image Sensing Self-check Quiz
Started
- Ungraded Plugin: 3.2 A
Brief History of Imaging
16 min
- Practice Quiz: 3.2 A
Brief History of Imaging Self-check Quiz
Started
- Reading: 3.3 Video
Correction
10 min
- Ungraded Plugin: 3.3
Types of Image Sensors
11 min
- Practice Quiz: 3.3
Types of Image Sensors Self-check Quiz
Started
- Reading: 3.4 Video
Correction
10 min
- Ungraded Plugin: 3.4
Resolution, Noise, Dynamic Range
14 min
- Practice Quiz: 3.4
Resolution, Noise and Dynamic Range Self-check Quiz
15 min
- Ungraded Plugin: 3.5
Sensing Color
19 min
- Reading: 3.5
Sensing Color Supplementary Reading
10 min
- Practice Quiz: 3.5
Sensing Color Self-check Quiz
10 min
- Ungraded Plugin: 3.6
Camera Response and HDR Imaging
17 min
- Practice Quiz: 3.6
Camera Response and HDR Imaging Self-check Quiz
10 min
- Ungraded Plugin: 3.7
Nature's Image Sensors
8 min
- Discussion Prompt: Week 3
Designing a Camera Lens
10 min
- Discussion Prompt: Week 3
Questions and Feedback

Week 3 Quiz

🎉 Congratulations! You passed!

Grade received 100% To pass 50% or higher

Go to next item

3.4 Resolution, Noise and Dynamic Range Self-check Quiz

1. Which of the following types of noise in image sensors is not scene independent?

1 / 1 point

Photon Shot Noise

Readout Noise

Dark Current Noise

Fixed Pattern Noise

Correct

Submit your assignment

Try again

coach

Ready to review what you've learned before taking the quiz? I'm here to help.

Help me practice

Let's chat

Your grade

100%

View Feedback

We keep your highest score

2. Photon shot noise can always be approximated to a Gaussian Distribution.

1 / 1 point

True

False

Like

Dislike

Report an issue

Correct

Photon shot noise can only be approximated to a Gaussian distribution for $k \gg 10$. Keep in mind that this is just an approximation. The Poisson distribution starts to "approximate" the Gaussian distribution when $k \gg 10$, but it is still a Poisson distribution.

3. Which of the following is the main cause of fixed pattern noise?

1 / 1 point

Photons arrive at the image sensors at a random rate due to the quantum nature of light.

Due to errors in converting the number of electrons at the image sensor into voltage.

Silicon electrons get released not because of the energy of incoming photons but due to heat energy around the camera

Slight differences between the characteristics of different pixels, introduced during the manufacturing of the image sensor.

Correct

Fixed pattern noise is caused due to discrepancies in the manufacturing of pixels. There will no doubt be slight variations in certain properties of pixels, such as the depth of the potential well, the efficiency of conversion of photons to electrons etc. causing a noisy final image.

4. Which of the following has the highest dynamic range?

1 / 1 point

Digital Video

Digital Camera

Human Eye

HDR Display

Correct

Dynamic range is the ratio of the maximum possible photon energy detectable (maximum brightness detectable) to the minimum possible photon energy detectable by a sensor. The human eye has a dynamic range of 120 dB, and can detect a level of brightness 5 times higher than the maximum brightness detectable by a HDR Display.

https://www.coursera.org/learn/cameraandimaging/quiz/SFYBI3-4-resolution-noise-and-dynamic-range-self-check-quiz?attempt=1&redirectToCover=true

1/1